

Notes: Lynch & Musto (JF, 2003)

How Investors Interpret Past Fund Returns

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1 Big picture

- **Question.** Why is the relation between mutual fund flows and past fund performance convex—that is, why do investors react strongly to good performance but appear relatively insensitive to differences among bad performers?
- **Core answer.** Past bad performance is not interpreted simply as a persistent signal of low skill. It also increases the probability that the investment advisor replaces the manager or abandons the strategy that produced the bad outcome.
- **Key mechanism.** The investment advisor holds an abandonment option. If performance is below a threshold R^* , the advisor replaces the old strategy. Once investors expect replacement, a bad return and a very bad return carry similar information about next-period performance.
- **Model implication.** Fund flows are a convex function of past returns. The slope of flows with respect to performance is low below the replacement threshold and high above the threshold.
- **Empirical implementation.** The paper uses daily mutual fund returns from Micropal, CRSP manager-change dates, Carhart four-factor loadings, and fund flow data to construct three proxies for strategy change.
- **Strategy-change proxies.** The proxies are: average absolute loading change, a top-quartile loading-change dummy, and a manager-change dummy.
- **Main empirical result 1.** Strategy-change proxies occur more often after bad performance.
- **Main empirical result 2.** For bad performers, flows are less sensitive to current performance when strategy-change proxies indicate replacement.
- **Main empirical result 3.** Future performance is less sensitive to bad current performance after manager change; the analogous evidence is weaker for factor-loading changes.
- **Economic takeaway.** Convex fund flows need not reflect irrationality or purely convex compensation. They can arise because rational investors understand that poor-performing strategies are often discarded.

2 Incremental contribution of the paper

- **Before this paper.** Prior work documented the convex flow-performance relation and performance persistence, often interpreting them through investor learning, marketing, convex manager incentives, or return-chasing.
- **What this paper adds.** Lynch and Musto add a simple but powerful state-dependence: the organizational technology inside the fund can change after poor outcomes. A fund's past return is therefore not only a signal about manager skill; it is also a signal about whether the same manager or strategy will continue.
- **New economic mechanism.** The paper converts the flow-performance puzzle into a real-options problem. Poor performance triggers abandonment, so investors rationally discount the information content of bad performance.
- **New empirical dimension.** Instead of only studying flows and returns, the paper tests the model using direct and indirect proxies for strategy replacement: manager turnover and changes in factor loadings.
- **Contribution relative to Heinkel-Stoughton.** The paper imports the manager-retention threshold logic into open-end mutual funds and uses risk-averse investor allocation to derive predictions for dollar flows.
- **Contribution relative to fund-flow studies.** The paper explains why the flow-performance relation is flatter among bad performers without assuming that investors ignore information. The return itself tells investors whether the production method is likely to persist.

- **Contribution relative to performance-persistence studies.** The paper links the convexity of fund flows to the convexity of persistence: when poor performance triggers strategy change, poor past returns become less predictive of future returns.
- **Why it matters.** The paper changes the interpretation of past returns. Investors are not necessarily extrapolating naively; they may be conditioning on endogenous replacement decisions inside the fund family.

3 Data and measurement

3.1 Mutual fund sample

- The empirical sample combines daily returns from Micropal with annual mutual fund data from CRSP and factor returns.
- The main loading-change tests use years 1985–1995 and require daily fund-return data in years $y - 1$, y , and $y + 1$.
- The manager-change tests use the overlapping period in which CRSP records manager-change dates.
- The sample focuses on equity fund types in the Micropal universe: Aggressive Growth, Growth and Income, and Long-Term Capital Gains.

Interpretation: The empirical design needs daily returns because strategy change is proxied through changes in factor loadings. Annual data alone cannot estimate the relevant Carhart loadings precisely.

3.2 Performance measures

- The paper estimates Carhart four-factor alphas using daily fund returns.
- Absolute performance is denoted $\alpha_{i,y}$.
- Group-adjusted performance is denoted $R\alpha_{i,y}$ and subtracts the average alpha of the fund’s category.
- Bad performance is defined in two ways: bottom-quartile performance or negative performance.

Interpretation: The two performance measures separate an absolute “did the fund earn alpha?” criterion from a relative “did it underperform comparable funds?” criterion.

3.3 Fund flows

Fund flows are measured as new dollar flows net of mechanical growth in existing assets:

$$DFL_{i,t} = TNA_{i,t} - TNA_{i,t-1}(1 + r_{i,t}),$$

where $TNA_{i,t}$ is total net assets and $r_{i,t}$ is the total return.

Interpretation: This is a dollar-flow measure, not a percentage-flow measure. It asks how much new money investors add or withdraw after adjusting for fund returns.

3.4 Strategy-change proxies

The paper uses three proxies for the unobserved decision to change strategy.

- **LDEL:** average absolute change in Carhart factor loadings from year $y - 1$ to $y + 1$.
- **LCOR:** LDEL scaled by standard errors of the estimated factor-loading changes.
- **QLCH:** an indicator equal to one if LDEL is in the top quartile for the fund's category and year.
- **QMCH:** an indicator equal to one if CRSP records a manager change during the relevant window.

Interpretation: LDEL and QLCH can detect style or algorithm changes even without a formal manager replacement. QMCH is cleaner but narrower because a strategy can change without a named manager change.

4 Models and empirical specifications

4.1 Fund return decomposition

The model decomposes the first-period return into ability, strategy, and period-specific components:

$$r_1 = \mu_0 + \varepsilon_a + \varepsilon_{s,1} + \varepsilon_{p,1}.$$

If the advisor keeps the old strategy, second-period return is

$$r_2 = \mu_0 + \varepsilon_a + \varepsilon_{s,1} + \varepsilon_{p,2}.$$

If the advisor changes strategy, second-period return is

$$r_2 = \mu_0 + \varepsilon_a + \varepsilon_{s,2} + \varepsilon_{p,2}.$$

Interpretation: The first-period return contains one component that always persists, one that persists only if the strategy is retained, and one that never persists. The strategy-specific component is what creates the option to abandon.

4.2 Initial investor allocation

Investor 0 chooses a dollar allocation to the fund based on the mean and variance of net returns. In the paper's CARA-normal setup, the initial allocation has the standard mean-variance form:

$$I_0 = \frac{(1 - \delta)\mu_0 - \delta}{\alpha(1 - \delta)^2(\sigma_a^2 + \sigma_s^2 + \sigma_p^2)}.$$

Interpretation: The exact expression matters less than the structure: risk-averse investors allocate more when expected performance is higher and less when uncertainty or fees are higher.

4.3 Advisor replacement threshold

The advisor compares the Sharpe ratio from keeping the old strategy to the Sharpe ratio from replacing it. There exists a cutoff R^* such that:

$$\text{keep old strategy} \iff r_1 > R^*.$$

Interpretation: A bad return has two opposing informational effects. It lowers beliefs about the old strategy, but it also makes replacement more likely. Below R^* , the future no longer depends strongly on just how bad the old strategy was.

4.4 Fund-flow function

Net fund flows are

$$F(r_1) = I_1(r_1) - I_0(1 - \delta)(1 + r_1).$$

The key theoretical prediction is that $F(r_1)$ has a kink at the strategy-replacement threshold R^* .

Interpretation: This is the core link from the replacement option to the empirical flow-performance relation. Above the threshold, better performance strongly predicts future performance because the strategy persists. Below the threshold, past performance is less predictive because the strategy is replaced.

4.5 Proposition 1: strategy-specific uncertainty creates convexity

For any $a < R^*$ and $b > R^*$,

$$F'(b) - F'(a) > 0 \quad \text{if strategy-specific uncertainty is positive.}$$

The kink is larger when strategy-specific uncertainty is larger.

Interpretation: Convex flows are not a mechanical feature of mutual funds; they come from the combination of uncertainty about strategy quality and the ability to abandon poor strategies.

4.6 Proposition 2: fund-specific ability changes the level of sensitivity

Setting uncertainty about fund-specific ability to zero does not eliminate the kink. Instead, fund-specific uncertainty raises the slope of the flow-performance relation both above and below the kink.

Interpretation: Uncertainty about persistent fund ability makes all past returns more informative, whereas uncertainty about the old strategy creates asymmetry around the replacement threshold.

4.7 Empirical regression for flows and future performance

The main piecewise-linear specification is

$$Y_{i,y+1} = b_0 + b_1 \text{proxy}_{i,y} + b_2 (\text{perf}_{i,y})^+ + b_3 \text{proxy}_{i,y} (\text{perf}_{i,y})^+ + b_4 (\text{perf}_{i,y})^- + b_5 \text{proxy}_{i,y} (\text{perf}_{i,y})^- + \varepsilon_{i,y+1},$$

where Y is either future dollar flow or future performance, and *proxy* is LDEL, QLCH, or QMCH.

Interpretation: The theory predicts $b_5 < 0$: among bad performers, the relation between current performance and future flow/performance should be flatter when the fund appears to have changed strategy.

5 Identification and empirical design

5.1 What identifies the empirical test

- The model predicts changes in both the incidence of strategy replacement and the slope of future outcomes with respect to past performance.

- The tests compare poor and good performers using independently constructed proxies for strategy change.
- The strongest evidence comes from consistency across three different proxies: continuous loading change, top-quartile loading-change indicator, and manager turnover.

Interpretation: The paper is not a randomized causal design. Its credibility comes from testing several model-specific predictions that would be hard to explain with a simple performance-chasing story.

5.2 Why the proxy structure matters

- Manager changes are direct but incomplete, because strategies can change without a recorded manager change.
- Loading changes are broader but noisier, because estimated factor loadings can move for mechanical or measurement-error reasons.
- LCOR reduces measurement-error concerns by scaling loading changes by standard errors.
- QLCH transforms the noisy continuous measure into a categorical high-change proxy.

Interpretation: Agreement across proxy types is important because each proxy has different errors. Evidence from all three strengthens the interpretation that investors condition on strategy replacement.

5.3 What the paper cannot fully prove

- The replacement decision itself is not fully observed for all forms of strategy change.
- Factor-loading changes are noisy measures of strategy replacement.
- Manager changes can be misdated or missing in CRSP snapshots.
- The paper does not estimate a fully structural model of investor demand and advisor replacement costs.
- It cannot explain why the very worst funds still receive some investment; it explicitly treats that as a separate clientele or behavioral/institutional issue.

Interpretation: The paper's main claim is not that every dollar of mutual fund flows is rational. It is that the broad convexity of flows and persistence can be explained by a rational interpretation of endogenous strategy replacement.

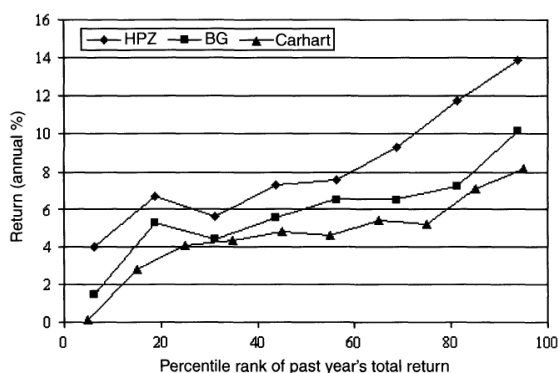
6 Tables and figures

6.1 Figure 1: Mean future returns net of risk-free rate and Figure 2: Mean future returns net of market risk

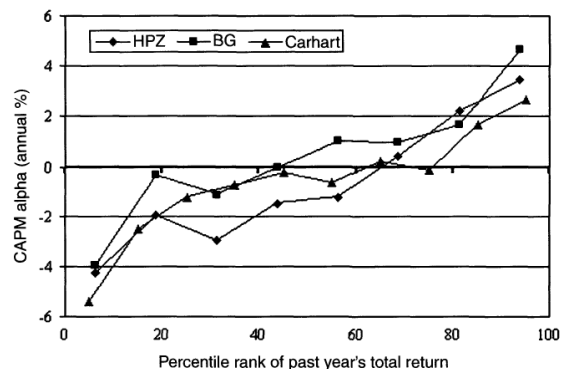
Read:

- Figure 1 plots future excess returns against percentile ranks of past total returns using results from Hendricks, Patel, and Zeckhauser; Brown and Goetzmann; and Carhart.
- Future returns rise with past returns, but the slope is not symmetric. Performance differences among good funds matter more than performance differences among bad funds.
- Figure 2 applies market-risk adjustment and shows the same qualitative pattern more clearly.
- Bad past performers generally have low future alpha, but the future-return curve is relatively flat over much of the lower-performance region.

Economic message: The empirical persistence relation already looks convex. This makes convex flows less puzzling: investors may rationally respond more strongly where past performance is more predictive of future performance.



(a) Figure 1: Mean future returns net of risk-free rate



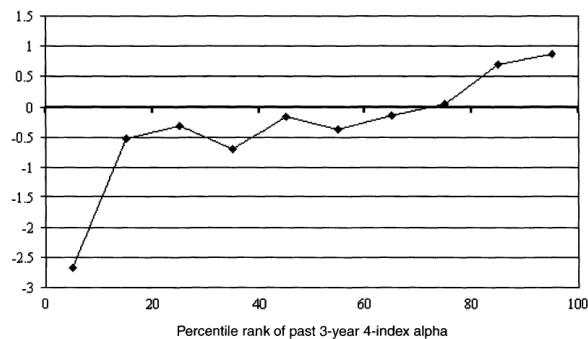
(b) Figure 2: Mean future returns net of market risk

6.2 Figure 3: Relationship between past and future monthly four-index alphas and Figure 4: Fund flows as a function of past return

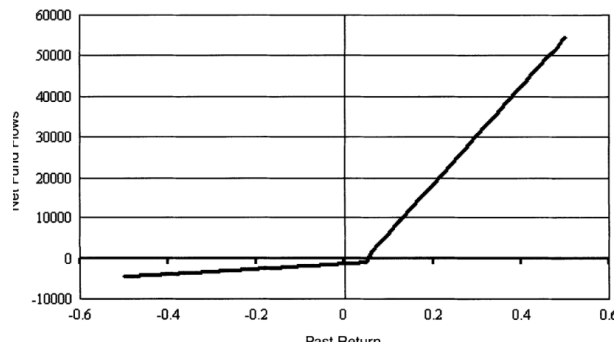
Read:

- Figure 3 reproduces Gruber's four-index alpha persistence pattern. Very poor past alphas do not map linearly into proportionally worse future alphas.
- The curve becomes more positive toward the high-performance region, consistent with stronger persistence when strategies are retained.
- Figure 4 displays the model-generated fund-flow function.
- Flows are only mildly responsive to performance below R^* , but much more responsive above R^* .
- The kink at R^* is the model's signature prediction: replacement flattens the bad-performance region.

Economic message: Figure 3 motivates the model, and Figure 4 shows the mechanism. The model turns a descriptive convexity in flows into a structural implication of strategy abandonment.



(a) Figure 3: Past and future four-index alpha



(b) Figure 4: Model-implied fund flows

6.3 Figure 5: Effect of strategy-specific portion on fund-flow response

Read:

- Figure 5 varies the variance of the strategy-specific component σ_s^2 .
- Higher strategy-specific uncertainty makes the flow-performance relation more convex.
- Below R^* , investors know the old strategy is likely to be abandoned, so additional bad news matters less.
- Above R^* , good performance makes persistence more likely and therefore has more predictive content.

Economic message: The figure distinguishes the key source of convexity from generic fund uncertainty. It is uncertainty about the replaceable strategy, not simply uncertainty about fund ability, that creates the kink.

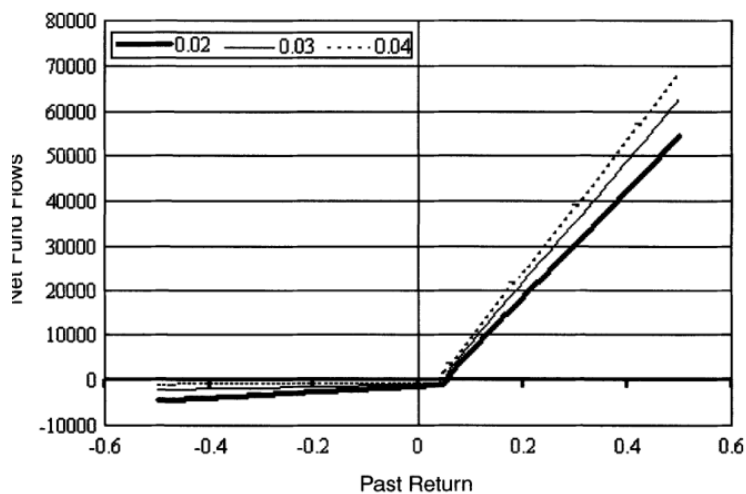


Figure 5: Effect of strategy-specific portion on fund-flow response

6.4 Table I: Loading-Change Results

Read:

- Panel A shows that the bottom performance quartile has higher average loading change. For group-adjusted alpha, LDEL is 0.204 in the bottom quartile versus 0.169 for the top three quartiles, with $p < 0.001$.

- LCOR also indicates more strategy movement for poor performers. For group-adjusted alpha, the bottom quartile LCOR is 1.931 versus 1.814 for the top three quartiles.
- Panel B uses continuous LDEL as the strategy-change proxy in piecewise regressions.
- For flows, the interaction with negative performance is negative and statistically meaningful. With group-adjusted alpha, the coefficient on $LDEL \times (perf)^-$ is -1799 with a t -statistic of -3.20; with alpha, it is -1905 with a t -statistic of -3.31.
- Future performance results are weaker for loading changes, consistent with performance being noisier than flows.

Economic message: Poor performers appear more likely to change factor exposures, and those changes flatten the relation between bad performance and future flows. This supports the replacement-option mechanism.

Table I
Loading-Change Results

For y from 1985 to 1995 for each fund i in the Micropal universe, daily fund returns are regressed on the simultaneous returns of the Carhart (1997) four factors for each year from $y-1$ to $y+1$ with at least 20 daily fund returns. The fitted coefficients are an intercept α_{it} and estimated factor loadings $b_{RMRF, it}$, $b_{SMB, it}$, $b_{HML, it}$, and $b_{MOM, it}$ for $t = y-1, y$, and $y+1$. The initial sample of 6,243 fund observations i, y are those with loadings data for years $y-1, y$, and $y+1$. $R\alpha_{it}$ is α_{it} minus the average α_{it} of the funds of i 's type (Aggressive Growth, Growth and Income, and Long-Term Capital Gains) in the Micropal universe with a usable loading observation in year t . $LDEL_{i, y}$ is the average absolute change of fund i 's factor loadings from $y-1$ to $y+1$ while $LCOR_{i, y}$ is $LDEL_{i, y}$ scaled by an estimate of its standard error. The fund flow variable used is $DFL_{i, y} \equiv (TNA_{i, y} - TNA_{i, y-1}(1 + r_{i, y}))$ where $TNA_{i, t}$ is the total net assets of fund i at the end of year t , and $r_{i, t}$ is the total return of fund i in year t . Any fund observation i, y without a value for $DFL_{i, y+1}$ is omitted: 283 observations are omitted for this reason, leaving 5,960 observations. Panel A reports average $LDEL_{i, y}$ and $LCOR_{i, y}$ among the funds with the indicated performance. The Quartiles p -value column reports p -values for a t -test of a higher average for the bottom than top three performance quartiles while the Sign Groups p -value column reports p -values for a t -test of a higher average for the <0 than the >0 performance groups. Panel B reports results for regressions of year- $(y+1)$ flow or year- $(y+1)$ performance on year- y performance, allowing for a kink in the regression line at zero year- y performance:

$$Y_{i, y+1} = b_0 + b_1 LDEL_{i, y} + b_2 (perf_{i, y})^+ + b_3 (LDEL_{i, y})(perf_{i, y})^+ + b_4 (perf_{i, y})^- + b_5 (LDEL_{i, y})(perf_{i, y})^- + \epsilon_{i, y+1}$$

where $(\cdot)^+ \equiv \max(0, \cdot)$ and $(\cdot)^- \equiv \min(0, \cdot)$. $LDEL_{i, y}$ is included as an independent variable and is allowed to interact both with negative and positive year- y performance. T -statistics based on White standard errors are below, in italics.

Panel A: Average $LDEL_{i, y}$ and $LCOR_{i, y}$ for Subsamples Formed on Year- y Performance (in %)

perf	All	Quartiles					Sign Groups		
		Bottom	3rd	2nd	Top	Top 3	p -val	<0	>0
$LDEL_{i, y}$									
R α	0.177	0.204	0.169	0.154	0.183	0.169	<0.001	0.183	0.171
α		0.205	0.163	0.150	0.191	0.169	<0.001	0.175	0.183
$LCOR_{i, y}$									
R α	1.841	1.931	1.825	1.809	1.812	1.814	0.001	1.874	1.804
α		1.894	1.847	1.758	1.866	1.824	0.016	1.834	1.854

Panel B: Regressions of Year- $(y+1)$ Flow or Year- $(y+1)$ Performance on Year- y Performance

Y	perf	b_0	b_1	b_2	b_3	b_4	b_5
DFL	R α	34.49	-21.70	2779	-1551	1328	-1799
		<i>5.03</i>	<i>-0.93</i>	<i>5.38</i>	<i>-0.93</i>	<i>6.80</i>	<i>-3.20</i>
	α	54.66	-35.92	1185	279	1488	-1905
perf	R α	<i>7.24</i>	<i>-1.56</i>	<i>2.37</i>	<i>0.19</i>	<i>6.86</i>	<i>-3.31</i>
		-0.00	-0.005	0.107	-0.246	0.270	0.192
	α	-0.29	-1.12	1.76	-0.97	5.22	0.92
		-0.01	-0.007	0.113	-0.216	0.323	0.107
		<i>-8.66</i>	<i>-1.63</i>	<i>1.41</i>	<i>-0.66</i>	<i>7.42</i>	<i>0.57</i>

Table I: Loading-Change Results

Table II
Loading-Change Dummy Results

For y from 1985 to 1995 for each fund i in the Micropal universe, daily fund returns are regressed on the simultaneous returns of the Carhart (1997) four factors for each year from $y-1$ to $y+1$ with at least 20 daily fund returns. The fitted coefficients are an intercept α_{it} and estimated factor loadings $b_{RMRF, it}$, $b_{SMB, it}$, $b_{HML, it}$, and $b_{MOM, it}$ for $t = y-1, y$, and $y+1$. The initial sample of 6,243 fund observations i, y are those with loadings data for years $y-1, y$, and $y+1$. $R\alpha_{it}$ is α_{it} minus the average α_{it} of the funds of i 's type (Aggressive Growth, Growth and Income, and Long-Term Capital Gains) in the Micropal universe with a usable loading observation in year t . $LDEL_{i, y}$ is the average absolute change of fund i 's factor loadings from $y-1$ to $y+1$. The fund flow variable used is $DFL_{i, y} \equiv (TNA_{i, y} - TNA_{i, y-1}(1 + r_{i, y}))$ where $TNA_{i, t}$ is the total net assets of fund i at the end of year t , and $r_{i, t}$ is the total return of fund i in year t . Any fund observation i, y without a value for $DFL_{i, y+1}$ is omitted: 283 observations are omitted for this reason, leaving 5,960 observations. $QLCH_{i, y}$ is 1 if $LDEL_{i, y}$ is in the top $LDEL$ quartile in year y for the funds in i 's category included in the sample, and 0 otherwise. Panel A reports the incidence of $QLCH_{i, y} = 1$ among the funds with the indicated performance. The Quartiles p -value column reports p -values for χ^2 tests of equal proportions across the bottom and top three performance quartiles while the Sign Groups p -value column reports p -values for χ^2 tests of equal proportions across the <0 and >0 performance groups. Panel B reports results for regressions of year- $(y+1)$ flow or year- $(y+1)$ performance on year- y performance, allowing for a kink in the regression line at zero year- y performance:

$$Y_{i, y+1} = b_0 + b_1 QLCH_{i, y} + b_2 (perf_{i, y})^+ + b_3 (QLCH_{i, y})(perf_{i, y})^+ + b_4 (perf_{i, y})^- + b_5 (QLCH_{i, y})(perf_{i, y})^- + \epsilon_{i, y+1}$$

where $(\cdot)^+ \equiv \max(0, \cdot)$ and $(\cdot)^- \equiv \min(0, \cdot)$. The $QLCH_{i, y}$ dummy is used in the regressions to estimate two separate regression lines, one for funds with $QLCH_{i, y} = 1$ and another for funds with $QLCH_{i, y} = 0$. T -statistics based on White standard errors are below, in italics.

Panel A: Proportion of Funds with $QLCH_{i, y} = 1$ for Subsamples Formed on Year- y Performance (in %)

perf	All	Quartiles					Sign Groups		
		Bottom	3rd	2nd	Top	Top 3	p -val	<0	>0
R α	25.17	33.75	22.27	18.49	26.51	22.46	<0.001	26.53	23.60
α		27.50	23.55	19.23	30.40	24.42	0.019	25.33	24.85

Panel B: Regressions of Year- $(y+1)$ Flow or Year- $(y+1)$ Performance on Year- y Performance

Y	perf	b_0	b_1	b_2	b_3	b_4	b_5
DFL	R α	29.49	6.91	2741	-1023	1136	-470
		<i>5.20</i>	<i>0.63</i>	<i>6.03</i>	<i>-1.40</i>	<i>6.94</i>	<i>-2.06</i>
	α	47.42	3.13	1434	-491	1252	-452
perf	R α	<i>7.66</i>	<i>0.27</i>	<i>3.24</i>	<i>-0.75</i>	<i>7.00</i>	<i>-1.85</i>
		-0.00	-0.00	0.067	-0.077	0.299	0.051
	α	-0.86	-1.07	1.55	-0.74	8.06	0.66
		-0.01	-0.00	0.069	-0.064	0.358	-0.004
		<i>-12.84</i>	<i>-1.42</i>	<i>1.43</i>	<i>-0.55</i>	<i>11.14</i>	<i>-0.05</i>

Table II: Loading-Change Dummy Results

6.5 Table II: Loading-Change Dummy Results

Read:

- QLCH equals one for funds in the top quartile of loading changes within category and year.
- Panel A shows that top-quartile loading changes are more common after poor group-adjusted performance: 33.75 percent in the bottom quartile versus 22.46 percent in the top three quartiles.
- Using alpha, the bottom quartile also has a high loading-change incidence of 27.50 percent, and the quartile test is statistically significant.

- Panel B shows that the flow sensitivity to bad performance is lower when QLCH equals one. With group-adjusted alpha, the negative-performance interaction is -470 with a t -statistic of -2.06.
- As with Table I, future-performance results are less consistent than flow results.

Economic message: The dummy version confirms the same story in a simpler classification: unusually large changes in factor exposure are concentrated after poor performance and weaken the flow response to bad performance.

6.6 Table III: Manager-Change Results

Read:

- Panel A shows that manager changes are much more common among poor performers.
- For group-adjusted alpha, 13.45 percent of bottom-quartile funds have a manager change, compared with 7.68 percent for the top three quartiles.
- For negative versus positive group-adjusted performance, the manager-change rate is 10.23 percent versus 6.17 percent.
- Panel B shows the expected flattening of flow sensitivity among bad performers after manager change. With alpha, the negative-performance interaction is -621 with a t -statistic of -1.98.
- The future-performance tests are strongest for manager change: the negative-performance interaction is -0.222 for group-adjusted alpha and -0.197 for alpha, both with sizable negative t -statistics.

Economic message: Manager change is the cleanest proxy for strategy replacement. Table III shows both incidence and outcome evidence: bad performance predicts manager turnover, and after turnover bad performance is less predictive of flows and future performance.

6.7 Table BI: Signed Loading-Change Results

Read:

- Table BI examines signed changes in individual factor loadings to evaluate an alternative explanation: managers might shift factor exposures after poor performance to exploit factor premia rather than truly replacing strategies.
- The table considers all four factors together, all but RMRF, and individual RMRF, SMB, HML, and MOM loadings.
- The paper reports little evidence that poor performers systematically increase exposure to factor premia in a way that would explain the main results.
- In particular, some apparent patterns are driven by good performers reducing exposures rather than bad performers increasing them.

Economic message: The appendix table protects the interpretation of the loading-change proxies. The main findings are not simply a story of poor performers mechanically loading up on known factors to boost CAPM alpha.

7 Short summary

- Mutual fund flows are convex in past performance: investors respond much more strongly to good performance than to differences among bad returns.

Table III
Manager-Change Results

For y from 1991 to 1994 for each fund i in the Micropal universe, daily fund returns are regressed on the simultaneous returns of the Carhart (1997) four factors for each year from y to $y+1$ with at least 20 daily fund returns. The fitted coefficients are an intercept $\alpha_{i,t}$ and estimated factor loadings $b_{RMRF,i,t}$, $b_{SMB,i,t}$, $b_{HML,i,t}$, and $b_{MOM,i,t}$ for $t=y$ and $y+1$. The initial sample of 4,663 fund observations i,y are those with loadings data for years y and $y+1$. $Rz_{i,t}$ is $\alpha_{i,t}$ minus the average $\alpha_{i,t}$ of the funds of its type (Aggressive Growth, Growth and Income, and Long-Term Capital Gains) in the Micropal universe with a usable loading observation in year t . $QMCH_{i,y}$ is 1 if the most recent manager change date recorded on CRSP for year $y+1$ occurred either sometime during year y (if no month is given) or between July of year y and June of year $y+1$; $QMCH_{i,y}$ is 0 if the most recent manager change date lies outside these ranges. A $QMCH_{i,y}$ value of 1 is changed to 0 if: (a) year y is the commencement year for the fund as recorded on CRSP, (b) the most recent manager change date recorded on CRSP for year $y+1$ occurred sometime during year y , and (c) the first fund return on CRSP is not for a month prior to the most recent manager change date. The fund flow variable used is $DFL_{i,t} \equiv (TNA_{i,t} - TNA_{i,t-1}(1 + r_{i,t}))$ where $TNA_{i,t}$ is the total net assets of fund i at the end of year t , and $r_{i,t}$ is the total return of fund i in year t . Any fund observation i,y without a value for $QMCH_{i,y}$ or $DFL_{i,y+1}$ is omitted: 312 observations were omitted for these reasons, leaving 4,351 observations. Panel A reports the incidence of $QMCH_{i,y} = 1$ among the funds with the indicated performance. The Quartiles p -value column reports p -values for χ^2 tests of equal proportions across the bottom and top three performance quartiles while the Sign Groups p -value column reports p -values for χ^2 tests of equal proportions across the <0 and >0 performance groups. Panel B reports results for regressions of year- $(y+1)$ flow or year- $(y+1)$ performance on year- y performance, allowing for a kink in the regression line at zero year- y performance:

$$Y_{i,y+1} = b_0 + b_1 QLCH_{i,y} + b_2 (perf_{i,y})^+ + b_3 (QLCH_{i,y})(perf_{i,y})^+ + b_4 (perf_{i,y})^- + b_5 (QLCH_{i,y})(perf_{i,y})^- + e_{i,y+1}$$

where $(\cdot)^+ \equiv \max(0, \cdot)$ and $(\cdot)^- \equiv \min(0, \cdot)$. The $QMCH_{i,y}$ dummy is used in the regressions to estimate two separate regression lines, one for funds with $QMCH_{i,y} = 1$ and another for funds with $QMCH_{i,y} = 0$. T -statistics based on White standard errors are below, in italics.

Panel A: Proportion of Funds with $QMCH_{i,y} = 1$ for Subsamples Formed on Year- y Performance (in %)

perf	All	Quartiles					Sign Groups		
		Bottom	3rd	2nd	Top	Top 3	p -val	<0	>0
Rz	9.08	13.45	10.25	6.91	5.94	7.68	<0.001	10.23	6.17
α		11.87	11.63	7.01	5.96	8.18	<0.001	11.94	6.42

Panel B: Regressions of Year- $(y+1)$ Flow or Year- $(y+1)$ Performance on Year- y Performance

Y	perf	b_0	b_1	b_2	b_3	b_4	b_5
DFL	Rz	60.56	-22.28	592	-83	969	-451
		<i>10.30</i>	<i>-1.26</i>	<i>3.58</i>	<i>-0.18</i>	<i>4.69</i>	<i>-1.30</i>
	α	74.93	-29.89	343	94	1140	-621
perf	Rz	10.85	-1.72	2.34	0.20	5.62	-1.98
		-0.00	-0.00	0.043	0.056	0.285	-0.222
	α	-0.16	-1.97	2.06	0.93	3.99	-2.35
		-0.01	-0.00	-0.003	0.108	0.259	-0.197
		<i>-14.61</i>	<i>-2.23</i>	<i>-0.14</i>	<i>1.61</i>	<i>4.98</i>	<i>-2.60</i>

Table III: Manager-Change Results

Table BI
Signed Loading-Change Results

For y from 1986 to 1994 for all funds i with data from $y-1$ to $y+1$, daily fund returns are regressed on the simultaneous returns of the Carhart (1997) four factors for each year from $y-1$ to $y+1$. The fitted coefficients are an intercept $\alpha_{i,t}$ and estimated factor loadings $b_{RMRF,i,t}$, $b_{SMB,i,t}$, $b_{HML,i,t}$, and $b_{MOM,i,t}$ for $t=y-1$, y , and $y+1$. $DEL_{F,i,y}$ is the average change of fund i 's factor loadings from $y-1$ to $y+1$ for the set of factors F . Six sets of factors are considered: all four factors (all four); all factors but RMRF (not RMRF); and each factor individually. For $j=2, 3$, and 4, $Q_{j,i,y}$ is 1 if $\alpha_{i,y}$ is in the j th quartile in year y for i 's category, and 0 otherwise. Regressions are run for the entire sample and have 6243 observations:

$$DEL_{F,i,y} = v_{F1} + \phi_{F2} Q_{2,i,y} + \phi_{F3} Q_{3,i,y} + \phi_{F4} Q_{4,i,y} + e_{F,i,y}$$

For $j=2, 3$, and 4, the regression coefficient ϕ_{Fj} measures the difference between the average factor change for performance (α) quartiles j and 1, while the intercept v_{F1} measures the average factor change for the bottom (first) performance quartile. Two hypotheses are tested: average factor change for quartile 1 equals the average factor change across the other three quartiles: $(\phi_{F4} + \phi_{F3} + \phi_{F2})/3 = 0$; and average factor change across all four quartiles equals zero: $v_{F1} + (\phi_{F4} + \phi_{F3} + \phi_{F2})/4 = 0$. T -statistics based on White standard errors are in bold.

F (Set of Factors)	v_{F1}	ϕ_{F2}	ϕ_{F3}	ϕ_{F4}	$(\phi_{F4} + \phi_{F3} + \phi_{F2})/3$	$v_{F1} + (\phi_{F4} + \phi_{F3} + \phi_{F2})/4$
All four	-0.010	-0.007	0.001	0.005	-0.001	-0.010
	-1.61	-0.91	0.08	0.63	-0.10	-4.18
Not RMRF	-0.004	-0.005	-0.000	0.007	0.001	-0.003
	-0.58	-0.62	-0.03	0.91	0.08	-1.26
RMRF	-0.027	-0.012	0.003	-0.003	-0.004	-0.030
	-3.96	-1.40	0.34	-0.37	-0.56	-10.33
SMB	0.011	-0.017	-0.004	0.005	-0.005	0.007
	1.04	-1.32	-0.38	0.40	-0.49	1.65
HML	-0.019	0.005	0.014	0.056	0.025	-0.000
	-2.12	0.44	1.25	4.34	2.46	-0.09
MOM	-0.003	-0.005	-0.010	-0.039	-0.018	-0.016
	-0.34	-0.44	-1.10	-3.79	-2.11	-4.78

We now turn to the results for the individual loading changes which are contained in the last four rows of each panel. The last column of Panel A shows that only the SMB loading has increased over the sample period, though not significantly, while both the market beta and the momentum loading have actually decreased significantly over the period. Comparing the loading change for the bottom performance quartile to the average for the other three, both panels indicate that funds increase their MOM loading and decrease their HML loading after near performance relative to average and good performers. However, with

Table BI: Signed Loading-Change Results

- The paper argues that investors interpret past returns jointly with the advisor’s option to replace poor strategies.
- A bad return signals low current strategy quality, but also signals that the strategy may not persist.
- The model generates a threshold R^* : above it the old strategy is retained; below it the old strategy is abandoned.
- This threshold creates a kink in the relation between flows and past performance.
- The kink is driven by uncertainty about the strategy-specific component of returns.
- The empirical analysis uses factor-loading changes and manager changes as proxies for strategy replacement.
- Poor performers are more likely to experience large loading changes and manager changes.
- For bad performers, flows are less sensitive to performance when replacement proxies indicate change.
- Future-performance evidence is strongest when replacement is measured by manager change.
- The paper provides a rational interpretation of a central mutual fund flow puzzle without denying that bottom-fund investors may be special or less performance-sensitive.

8 Conclusion

8.1 Core conclusion

The paper concludes that investors interpret past mutual fund returns through the lens of organizational adjustment. Performance does not simply reveal the permanent quality of the existing strategy. It also changes the probability that the strategy will continue. Because poor-performing strategies are more likely to be replaced, investors rationally treat differences among poor past returns as less informative about future performance.

8.2 Economic intuition

The intuition is the same as in any real-options problem. A firm that operates a bad technology abandons it; a mutual fund that runs a bad strategy or employs a bad manager can do the same. Once a strategy is abandoned, its past return no longer forecasts the future as strongly. Therefore, a poor return and a very poor return can both imply a similar future: a different strategy. Good performance, by contrast, makes persistence more likely, so differences among good returns remain informative.

8.3 Incremental contribution to mutual fund research

Relative to the existing literature on fund-flow convexity, the paper moves the mechanism inside the fund. It does not rely only on investor irrationality, marketing, convex compensation, or tournament incentives. It shows that the operating decision of the investment advisor—whether to keep or replace a strategy—is enough to produce a convex flow-performance relation. Relative to performance-persistence studies, it explains why persistence is itself weaker among poor performers: poor strategies are less likely to survive.

8.4 Implications for interpreting fund flows

Fund flows are not a pure measure of investors’ belief in historical performance. They are a measure of investors’ belief in the performance of the future fund, and the future fund may not be the same strategy or the same manager as the past fund. This distinction matters whenever organizations can revise production methods after bad outcomes.

8.5 Implications for empirical asset-management work

The paper suggests that empirical tests of persistence should condition on organizational change. A bad alpha followed by no manager change has different meaning from a bad alpha followed by a new manager or large factor-loading changes. Similarly, flow-performance regressions may miss important state dependence if they do not allow the predictive content of past returns to change after replacement.

8.6 Limitations and open questions

The paper does not directly observe every form of strategy replacement. Loading-change proxies are noisy, and manager changes capture only one kind of strategy change. The model also does not fully explain why some investors remain in the very worst funds; the paper leaves that issue to institutional frictions, behavioral biases, or disadvantaged-clientele explanations. A natural extension would estimate a structural model in which investors jointly infer skill, replacement probability, and replacement costs.

8.7 Takeaway

The lasting takeaway is that bad performance may be less informative precisely because it causes action. Investors are not only learning about a manager; they are also learning about whether that manager or strategy will be allowed to continue. This endogenous discontinuity in persistence provides a clean explanation for convex mutual fund flows.